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SUBJECT CODE NO: - RNP-8001**FACULTY OF SCIENCE AND TECHNOLOGY****F.E. (Group A & B) Mech/ Civil / CSE / AI & ML / EEE (NEP) Sem - I****Examination June/July 2025****Applied Mathematics-I****[Time: 3 Hours]****[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.no2, 3, 4, 5, 6 and 7

SECTION A**Q.1 Solve the following questions (Any Ten)****20**

- a) If $f(x) = e^x$, what is its Maclaurin's series expansion?
- a) $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$ b) $1 - x - \frac{x^2}{2!} - \frac{x^3}{3!} - \frac{x^4}{4!} + \dots$
- c) $x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$ d) $1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots$
- b) Find the Taylor series expansion of the function $\cosh(x)$ at $x = 0$.
- a) $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \infty$ b) $\frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \infty$
- c) $1 + \frac{x^2}{2!} + \frac{x^4}{4!} - \dots \infty$ d) $1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \infty$
- c) L'Hôpital's Rule can be used to resolve which of the following indeterminate forms?
- a) $0/0$ and ∞/∞ b) 1^∞ and 0^0
- c) $0 \times \infty$ and $\infty - \infty$ d) All of the above
- d) The series $\sum \frac{1}{n}$:
- a) Converges to a finite value. b) Diverges to infinity.
- c) Converges conditionally. d) Oscillates
- e) The stationary points of the function $f(x, y) = x^3 + y^3 - 3axy$ are
- a) $x = 0, y = a$ b) $x = a, y = 0$
- c) $x = a, y = a$ d) $x = a, y = -a$
- f) If $u = \log(x^3 + y^3 - x^2y - xy^2)$ then $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial y^2} + y^2 \frac{\partial^2 u}{\partial z^2}$ is equal to
- a) -9 b) -1 c) -3 d) 0

- g) If $x = uv$ & $y = \frac{u}{v}$ then JJ' is equal
 a) 0 b) 1 c) -1 d) 2
- h) The necessary & sufficient condition for D.E. to be exact D.E. is
 a) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ b) $\frac{\partial M}{\partial y} = -\frac{\partial N}{\partial x}$
 c) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$ d) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$
- i) D.E. of the form $\frac{dy}{dx} + P(x)y = Q(x)y^n$ is called as
 a) Linear D.E. LDE b) Exact D.E.
 c) Bernoulli's D.E. d) Reducible to LDE
- j) Newton's second law of motion if
 a) $F = mv \frac{dv}{dx}$ b) $F = v \frac{dv}{dx}$
 c) $F = m \frac{dv}{dx}$ d) $F = ma \frac{dv}{dx}$
- k) If u is homogeneous function of x, y of degree n then by Euler's thm $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ is equal to
 a) 0 b) 1 c) nu d) u
- l) If $x = e^\theta \cos \theta$ $y = e^\theta \sin \theta$ then $\frac{\partial(x,y)}{\partial(\theta, \phi)}$ is
 a) $x + y$ b) $x^2 + y^2$ c) e^θ d) 1
- m) Integrating factor of the D.E. $\frac{dy}{dx} + y \cot x = \sin^2 x$ is
 a) $\sin x$ b) $\cos x$ c) $\cot x$ d) $\operatorname{cosec} x$

SECTION B

Q.2 a) Expand $\log \sec x$ in descending powers of x 05

b) Evaluate: $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^2}}$ 05

Q.3 a) If $u = \log(x^2 + y^2)$ then prove that: $u_{xy} = u_{yx}$ 05

b) A rectangular box open at the top is to have volume 256 cubic feet, determine dimensions of the box required least 05

Q.4 a) Solve: $(x + 2y^3) \frac{dy}{dx} = y$ 05

b) A resistance of 100Ω , an inductance of 0.5 Henry is connected in series with a battery of 20 volts. Find the current in the circuit at $t = 0.5$ sec if $i = 0$ at $t = 0$ 05

Q.5 a) Expand e^x in powers of $(x - 2)$.

05

b) Test the convergence of series $\sum \frac{1}{\left(1+\frac{1}{n}\right)^{n^2}}$

05

Q.6 a) If $u = x + y + z$, $u^2v = y + z$, $u^2w = z$ then find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$

05

b) If $u = \sec^{-1}\left(\frac{x^3+y^3}{x+y}\right)$ then prove that $xu_x + yu_y = 2cotu$

05

Q.7 a) Solve: $\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$

05

b) $\frac{dy}{dx} - y \tan x = y^4 \sec x$

05